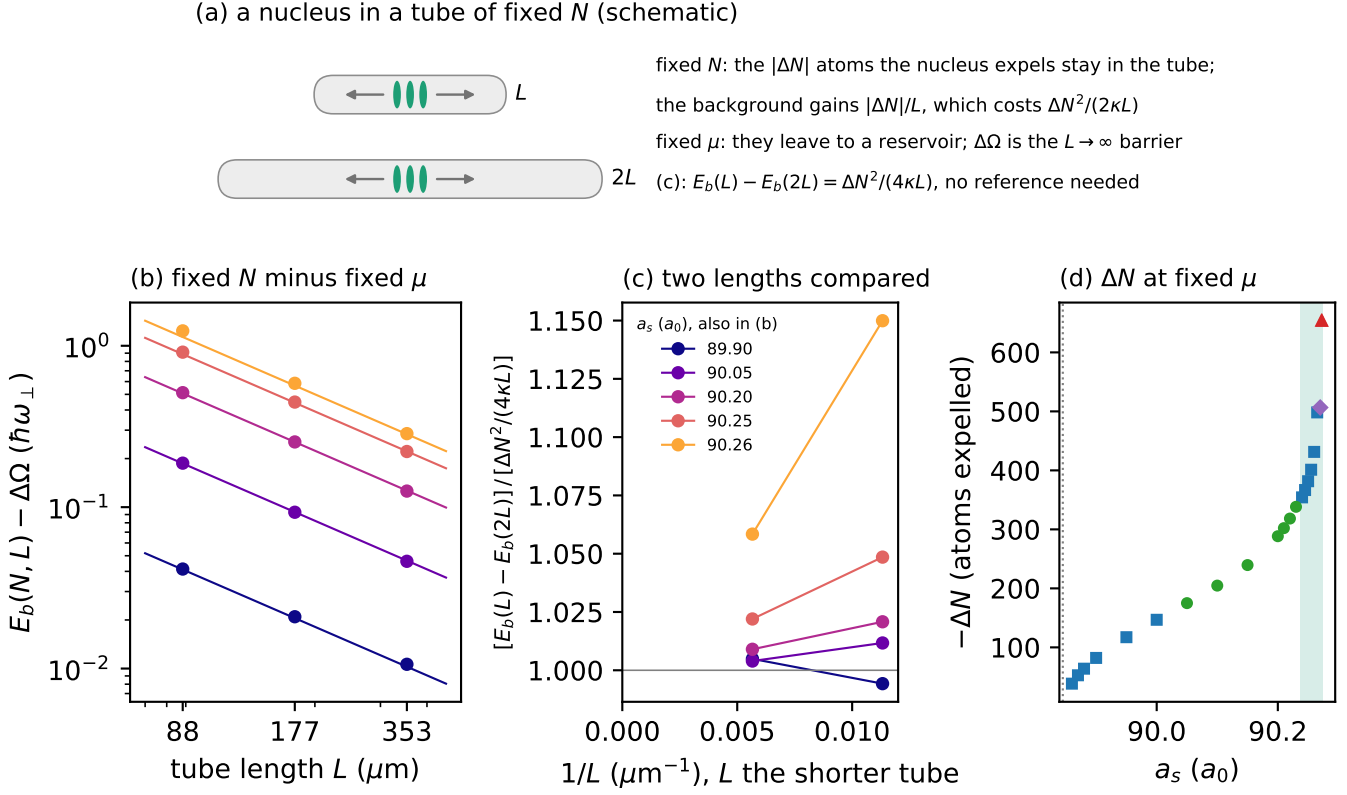
(a) a nucleus in a tube of fixed N (schematic)
L
$2L$
fixed N: the $|\Delta N|$ atoms the nucleus expels stay in the tube;
the background gains $|\Delta N|/L$, which costs $\Delta N^2/(2\kappa L)$
fixed μ: they leave to a reservoir; $\Delta\Omega$ is the $L \to \infty$ barrier
(c): $E_b(L) - E_b(2L) = \Delta N^2/(4\kappa L)$, no reference needed
(b) fixed N minus fixed μ
$E_b(N, L) - \Delta\Omega\ (\hbar\omega_\perp)$
10^0
10^{-1}
10^{-2}
88
177
353
tube length L (μm)
(c) two lengths compared
$[E_b(L) - E_b(2L)]/[\Delta N^2/(4\kappa L)]$
1.150
1.125
1.100
1.075
1.050
1.025
1.000
a_s (a_0), also in (b)
89.90
90.05
90.20
90.25
90.26
0.000
0.005
0.010
$1/L$ (μm^{-1}), L the shorter tube
(d) ΔN at fixed μ
$-\Delta N$ (atoms expelled)
600
500
400
300
200
100
90.0
90.2
a_s (a_0)